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Visual Servoing Using Sliding-Mode Control with Dynamic Compensation for UAVs' Tracking of Moving Targets

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Abstract: An Image-Based Visual Servoing Control (IBVS) structure for target tracking by Unmanned Aerial Vehicles (UAVs) is presented. The scheme contains two stages. The first one is a sliding-model controller (SMC) that allows one to track a target with a UAV; the control strategy is designed in the function of the image. The proposed SMC control strategy is commonly used in control systems that present high non-linearities and that are always exposed to external disturbances; these disturbances can be caused by environmental conditions or induced by the estimation of the position and/or velocity of the target to be tracked. In the second instance, a controller is placed to compensate the UAV dynamics; this is a controller that allows one to compensate the velocity errors that are produced by the dynamic effects of the UAV. In addition, the corresponding stability analysis of the sliding mode-based visual servo controller and the sliding mode dynamic compensation control is presented. The proposed control scheme employs the kinematics and dynamics of the robot by presenting a cascade control based on the same control strategy. In order to evaluate the proposed scheme for tracking moving targets, experimental tests are carried out in a semi-structured working environment with the hexarotor-type aerial robot. For detection and image processing, the Opencv C++ library is used; the data are published in an ROS topic at a frequency of 50 Hz. The robot controller is implemented in the mathematical software Matlab.

Keywords: visual servo control; sliding-model control; dynamic compensation; UAVs' tracking

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1. Introduction

Nowadays, robotics has developed great innovations, offering a variety of robots that perform different tasks in different environments. One of the robots that has evolved the most in recent years is the so-called Unmanned Aerial Vehicle (UAV), more commonly known as a drone. These types of robots are known for their great ability to move from one place to another, without the mobility restrictions commonly found in terrestrial mobile robots. They are used in various applications, such as inspection [1], agriculture [2], the transport of objects to inhospitable places [3,4], search and rescue operations [5,6], and aerial photogrammetry [7,8], among other applications. All these applications can be executed autonomously, semi-autonomously, or in a tele-operated way where the human being totally controls the robot. For the execution of autonomous tasks, it is necessary to establish a control algorithm that allows the robot to perform each task safely and in the shortest possible time [9,10].

Most control algorithms are based on the robot's model; therefore, it is essential to have a model that describes the characteristics and restrictions of the robot's movement, in order to guarantee the safety and efficiency of the UAV when performing the desired task. There are two types of models that are used to propose control schemes. In [11,12], the authors used the kinematic model of the robot to develop a control scheme that allows multiple tasks to be performed, such as positioning, trajectory tracking, and path following. To do so, the authors employed a unified control law that solves all three tasks by modifying only the controller references. Similarly, in [3], a controller for the transport of loads by two UAVs is presented, implementing a kinematic strategy that allows cooperation between the two robots. Another control scheme based on the kinematic model is presented in [13]. The authors used Lyapunov theory to design a nonlinear kinematic controller for path following in 3D space. The controller consists of two stages: the first is an external stage that is responsible for guiding the UAV towards the desired path, while the second is an internal stage that acts as an autopilot, stabilizing the nonlinear dynamics and following the commands generated by the external loop. As can be seen, with the kinematic model, it is possible to develop control strategies for multiple tasks, all without considering the robot's dynamic parameters. On the other hand, there are also approaches that use the dynamic model of the robot, with the aim of accounting for the robot's dynamic effects. This dynamic model represents the robot's movement as closely as possible to real-life behavior. In [14], the authors present an adaptive control of a UAV, with the aim of adjusting the dynamic parameters according to the configuration of the robot or the load that it carries; for the development, the authors used the dynamic model of the UAV according to forces and torques. On the other hand, in [15], the authors present the development of a nonlinear hybrid control system, where MPC control and fuzzy logic are combined. Similarly, the authors used a dynamic model that accounts for the robot's complex dynamics. They evaluated the robot's performance during flight and in the presence of external disturbances. In addition to these detailed examples, there are other works that present controllers based on robot dynamics, where specific tasks are carried out using the proposed strategies [16,17].

According to the aforementioned, there are multiple control strategies in the literature that involve the mathematical models of the robot, including strategies that combine both models. Specifically, they use the kinematic model to perform the task and, in sequence, a dynamic controller to compensate for disturbances such as wind and/or dynamic effects produced during task execution [11]. Most of the tasks that can be performed with these robots use sensors that provide feedback to the controller, such as position sensors, velocity sensors, and laser sensors, among others. However, for more sophisticated tasks such as aerial photography, bridge inspection, crop inspection, and the identification and rescue of people, it is necessary to have sensors that provide the necessary information to the robot, such as cameras [18]. Currently, most robots—mobile, terrestrial, aquatic, and space robots—use cameras that provide them with the ability to perceive their working environment [19,20]. Therefore, by having a camera, the robot improves its autonomy and can perform tasks accurately and reliably, even performing specific applications of tracking and searching for targets. In the military field, these robots are used for aerial surveillance, intelligence gathering, target identification, the identification of illegal mining, and the identification of areas used for drug trafficking, among other applications that put human beings at risk [16,17]. The ability to have vision in a UAV facilitates the coordination of rescue efforts, even leading to saving lives in danger.

In the literature, there are works related to visual servo control for UAVs to focus on the tracking or detection of targets. For example, in [21] they developed a UAV tracking controller to track moving targets, where the target performs unknown translational and rotational motions. The controller seeks to control both the relative position and orientation between the UAV and the target, ensuring that the target remains within the field of view of the UAV's on-board camera. They implement a predictive control strategy focused on predicting the future motion of the target and ensure that control actions can be performed by the UAV. Likewise, [22] presents a visual servo control by means of a propfunding algorithm, called Deep Q-Network (DQN), which focuses on adjusting the control gain in order to avoid the loss of the target's field of view (FOV). The controller is based on Image-Based Visual Servoing (IBVS), which uses feedback from the positions of the characteristic points in the image to generate the control commands. On the other hand, in [23], the authors proposed an Image-Based Visual Servo system for quadcopters; they realized five different designs of sliding surfaces, using analytical techniques and fuzzy logic. The visual servo system is only proposed in a simulated form. However, the strategy proposed by the authors only performs position control, i.e., it does not perform tracking; rather, it only maintains the image characteristics at the desired position in the image plane. As can be seen, there are works focused on visual servo control using different techniques. Consequently, in this work, we propose the visual servo control for the tracking of moving targets, maintaining an orientation relative to the target and a distance relative to it; by means of this we have a safe tracking of the moving target. In addition, since the target is mobile, a linear velocity estimator is used to guarantee the tracking by the UAV and to avoid the target leaving the robot's field of view (FOV). To track the target and to maintain the orientation and tracking distance, the redundancy of the system is used, where several tasks can be defined for the robot to accomplish. The expected contributions of this work are as follows:

- Visual servo control based on the sliding-mode technique for tracking moving targets;
- Control scheme based on secondary tasks, where target tracking is considered as the primary task, and distance and relative orientation are set as secondary tasks to be accomplished by the UAV;
- A dynamically compensated control scheme based on sliding mode, in which uncertainties in the modeling or disturbances external to the system are considered.

This document is organized into six sections, including the introduction. Section 2 describes the UAV models for the development of controllers. Section 3 details the camera model, describing the behavior of the target in the image plane. Section 4 presents the development of the controllers proposed in this work, together with their respective stability and robustness analysis. In Section 5, the results obtained from the proposed controller are presented, detailing the robot used in this work. In Section 6, the conclusions drawn from the experimental results of the proposed controller are shown. Finally, the terms and symbols used in this article are described in detail in the Table of Nomenclature, located in Appendix A.

2. Kinematic and Dynamic Model

2.1. Kinematic Model

The kinematics of the UAV are a vector describing the position and orientation of the robot within the fixed reference system, denoted by $\{\mathcal{F}\}$. The control point for the task is given by the position of the center of mass of the UAV $\boldsymbol{\xi}_{\mathbf{p}} \in \Re^3$ and $\xi_\psi \in \Re$ as orientation, such that

$$\boldsymbol{\xi}(t) = \begin{bmatrix} \boldsymbol{\xi}_{\mathbf{p}}(t) \\ \xi_\psi(t) \end{bmatrix} = \begin{bmatrix} \xi_x(t) \\ \xi_y(t) \\ \xi_z(t) \\ \xi_\psi(t) \end{bmatrix} \tag{1}$$

Now, the velocity of the defined operating point is given by the time derivative of (1), such that $\dot{\boldsymbol{\xi}}(t) = \frac{d}{dt}\boldsymbol{\xi}(t)$, which results in

$$\dot{\boldsymbol{\xi}}(t) = \begin{bmatrix} \dot{\boldsymbol{\xi}}_{\mathbf{p}}^T & \dot{\xi}_\psi \end{bmatrix}^T = \begin{bmatrix} \dot{\xi}_x & \dot{\xi}_y & \dot{\xi}_z & \dot{\xi}_\psi \end{bmatrix}^T \tag{2}$$

To obtain the relationship between the robot operating point or task velocities as a function of the robot joint velocities, we use the matrix that relates the moving reference frame of the robot $\{\mathcal{A}\}$ to the fixed reference frame $\{\mathcal{F}\}$ is used. The matrix is obtained by

$\mathbf{R}(\phi, \theta, \psi) = \mathbf{R}(x, \phi)\mathbf{R}(y, \theta)\mathbf{R}(z, \psi)$. Figure 1 shows the kinematic configuration of the UAV in the fixed reference system $\{\mathcal{F}\}$.

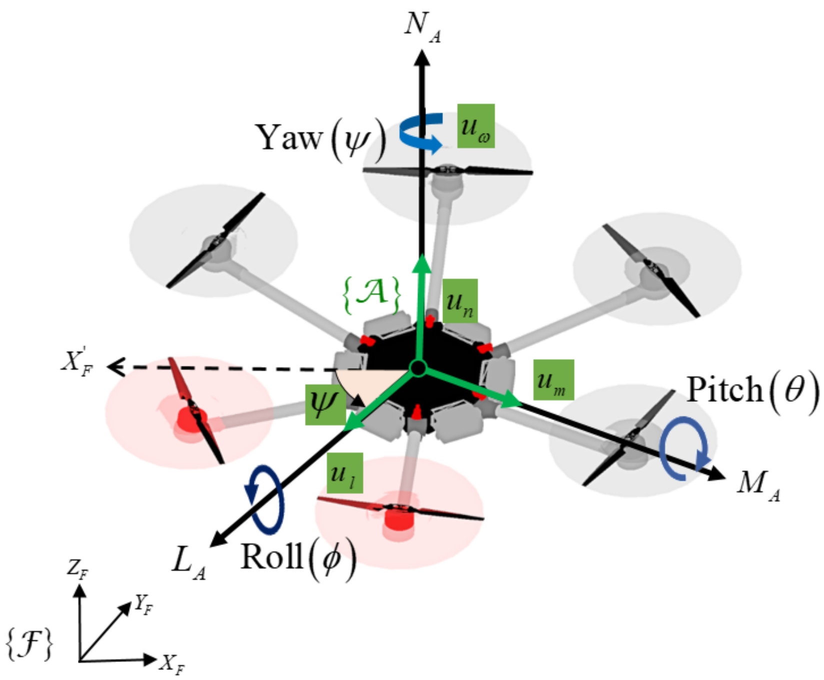


Figure 1. UAV kinematic configuration in the workspace $\{\mathcal{F}\}$, where X_F' is a line parallel to the axis-X_F of the fixed reference system; this allows one to identify the axis from which the UAV's orientation ψ is measured.

Assumption 1. *In this case, it can be observed that the roll and pitch angles of the UAV are not considered. This takes in consideration that the tasks defined for the robot are at low velocities, especially for drones used in service robotics. Therefore, the angular rotations of roll and pitch are very close to zero, and the matrices* $\mathbf{R}(x, \phi) = \mathbf{R}(y, \theta) = \mathbf{I}$ *result in identity matrices. Therefore,* $\mathbf{R}(\phi, \theta, \psi) = \mathbf{R}(\psi)$.

The aerial robot used in this work (Figure 1) has three linear velocities: u_l, u_m, and u_n, which allow the robot displacement in the working space $\{\mathcal{F}\}$, and an angular velocity u_ψ, which is the rotational velocity of the robot around the Z-axis. Therefore, the lineal velocity of the robot in relation to the maneuverability velocities are given by $\dot{\boldsymbol{\xi}}_\mathbf{p}(t) = \mathbf{R}(\psi)\begin{bmatrix} u_l(t) & u_m(t) & u_n(t) \end{bmatrix}^T$, while, for the rotation velocity, it is $\dot{\xi}_\psi(t) = u_n(t)$. Based on this, the velocity vector of the operational workspace is given by

$$\dot{\boldsymbol{\xi}}(t) = \overbrace{\begin{bmatrix} \mathbf{R}(\psi) & \mathbf{0}_{3\times 1} \\ \mathbf{0}_{1\times 3} & 1 \end{bmatrix}}^{\mathbf{J_A}(\psi)} \mathbf{u}(t) \tag{3}$$

where, $\mathbf{u}(t) = \begin{bmatrix} u_l & u_m & u_n & u_\psi \end{bmatrix}^T$ is the vector of maneuverability velocities of the robot and $\mathbf{J_A}(\psi) \in \Re^{4\times 4}$ is the matrix that relates maneuverability velocities and operational velocities, also known as the Jacobian of the system. The differential kinematics of the robot written in a compact form are $\dot{\boldsymbol{\xi}}(t) = \boldsymbol{J}_\mathbf{A}(\psi)\mathbf{u}(\mathbf{t})$, with

$$\begin{bmatrix} \dot{\xi}_x(t) \\ \dot{\xi}_y(t) \\ \dot{\xi}_z(t) \\ \dot{\xi}_\psi(t) \end{bmatrix} = \begin{bmatrix} \cos(\psi) & -\sin(\psi) & 0 & 0 \\ \cos(\psi) & \cos(\psi) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u_l \\ u_m \\ u_n \\ u_\psi \end{bmatrix} \tag{4}$$

2.2. Simplified Dynamic Model

The dynamic model used in this work is a simplified dynamic model, considering the dominant dynamics of the robot. The structure of the model is presented in [24]. However, the dynamic parameters corresponding to the robot need to be identified and validated. The dynamic model of the UAV is given by

$$\mathbf{u_{ref}}(t) = \mathbf{M}(\delta)\dot{\mathbf{u}}(t) + \mathbf{C}(\delta)\mathbf{u}(t) \tag{5}$$

where $\mathbf{u_{ref}}(t) = \begin{bmatrix} u_{l_{ref}} & u_{m_{ref}} & u_{n_{ref}} & u_{\psi_{ref}} \end{bmatrix}^T$ is the reference velocities vector; $\dot{\mathbf{u}}(t) \in \Re^4$ is the acceleration vector of the robot, defined by $\dot{\mathbf{u}}(t) = \begin{bmatrix} \dot{u}_l & \dot{u}_m & \dot{u}_n & \dot{u}_\psi \end{bmatrix}^T$; and $\mathbf{M}(\delta)$, $\mathbf{C}(\delta) \in \Re^{4\times4}$ are diagonal matrices that depend on the parameter vector of the dynamic model $\delta \in \Re^p$. The matrices are defined by

$$\mathbf{M}(\delta) = \begin{bmatrix} \delta_1 & 0 & 0 & 0 \\ 0 & \delta_2 & 0 & 0 \\ 0 & 0 & \delta_3 & 0 \\ 0 & 0 & 0 & \delta_4 \end{bmatrix} \quad \mathbf{C}(\delta) = \begin{bmatrix} \delta_5 & 0 & 0 & 0 \\ 0 & \delta_6 & 0 & 0 \\ 0 & 0 & \delta_7 & 0 \\ 0 & 0 & 0 & \delta_8 \end{bmatrix} \tag{6}$$

The dynamic model can also be expressed in parametric form as $\mathbf{u_{ref}} = \mathbf{D_P}(\mathbf{u}, \dot{\mathbf{u}})\delta$

$$\bar{\mathbf{u}}_r = \begin{bmatrix} \bar{\mathbf{u}}_{\mathbf{ref}}(t_1) \\ \bar{\mathbf{u}}_{\mathbf{ref}}(t_2) \\ \vdots \\ \bar{\mathbf{u}}_{\mathbf{ref}}(t_N) \end{bmatrix} = \begin{bmatrix} \mathbf{D_P}(t_1) \\ \mathbf{D_P}(t_2) \\ \vdots \\ \mathbf{D_P}(t_N) \end{bmatrix} \delta = \bar{\mathbf{D}}_{r\times p}\delta_p \quad r >> p \tag{7}$$

Now, let us consider the system of equations of the form $\bar{\mathbf{u}}_r = \bar{\mathbf{D}}\delta$, with $\bar{\mathbf{D}} \in \Re^{s\times p}$; then, the solution to this system by minimum squares is given by

$$\hat{\delta} = \left(\bar{\mathbf{D}}^T\bar{\mathbf{D}}\right)^{-1}\bar{\mathbf{D}}^T\bar{\mathbf{u}}_r \tag{8}$$

From the solution to (8), the estimated parameters are

$$\hat{\delta} = \begin{bmatrix} 0.9605 & 0.9489 & 0.2746 & 0.0617 & 0.6824 & 0.7163 & 0.9788 & 1.001 \end{bmatrix}^T \tag{9}$$

Assumption 2. *From Equation (5), we have a simplified model. It can be seen that the model does not consider modeling errors and/or disturbances, for which an additional vector is assumed to exist. The dynamic model that takes into account the modeling errors and/or disturbances of the UAV is expressed as follows:*

$$\mathbf{u_{ref}}(t) = \mathbf{M}(\delta)\dot{\mathbf{u}}(t) + \mathbf{C}(\delta)\mathbf{u}(t) + \boldsymbol{\zeta}(t) \tag{10}$$

where $\boldsymbol{\zeta}(t) \in \Re^4$ is the vector of perturbations and errors in the modeling, with $\|\boldsymbol{\zeta}\| \leq b$, i.e., it is a vector bounded by $b \in \Re^+$.

3. Differential Kinematics of the Target Image Feature

3.1. Considerations

Let $\mathcal{O}_\mathcal{F} = \{\mathcal{F}, X_F, Y_F, Z_F\}$ be the fixed reference frame; $\mathcal{O}_\mathcal{A} = \{\mathcal{A}, L_A, M_A, N_A\}$ be the UAV center-of-mass reference system; $\mathcal{O}_\mathcal{C} = \{\mathcal{C}, X_C, Y_C, Z_C\}$ be the reference frame of the camera mounted on the UAV; $\mathcal{O}_o = \{\mathcal{O}, X_o, Y_o, Z_o\}$ be the target reference system; $\mathcal{O}_\mathcal{I} = \{\mathcal{I}, U_I, V_I\}$ be the reference frame of the image plane, where the position of the target to track is projected; $^{\mathbf{f}}\mathbf{R_a} \in SO(3)$ be the rotation matrix of the UAV reference system with respect to the fixed system; $^{\mathbf{a}}\mathbf{R_c} \in SO(3)$ be the rotation matrix of the camera with respect to the reference frame of the UAV center of mass; and $^{\mathbf{c}}\mathbf{R_o} \in SO(3)$ be the rotation matrix of the object (target) with respect to the reference system of the camera. In addition, the position of the camera with respect to the center of mass of the UAV is denoted by

${}^{\mathbf{a}}\mathbf{p_c} \in \Re^3$; the position of the target of interest with respect to the fixed system is given by ${}^{\mathbf{f}}\mathbf{p_o} \in \Re^3$; and ${}^{\mathbf{c}}\mathbf{p_o} \in \Re^3$ is the position of the target with respect to the camera position. Figure 2 shows the configuration of the on-board camera system, which shows the UAV with the camera attached to the base of the UAV.

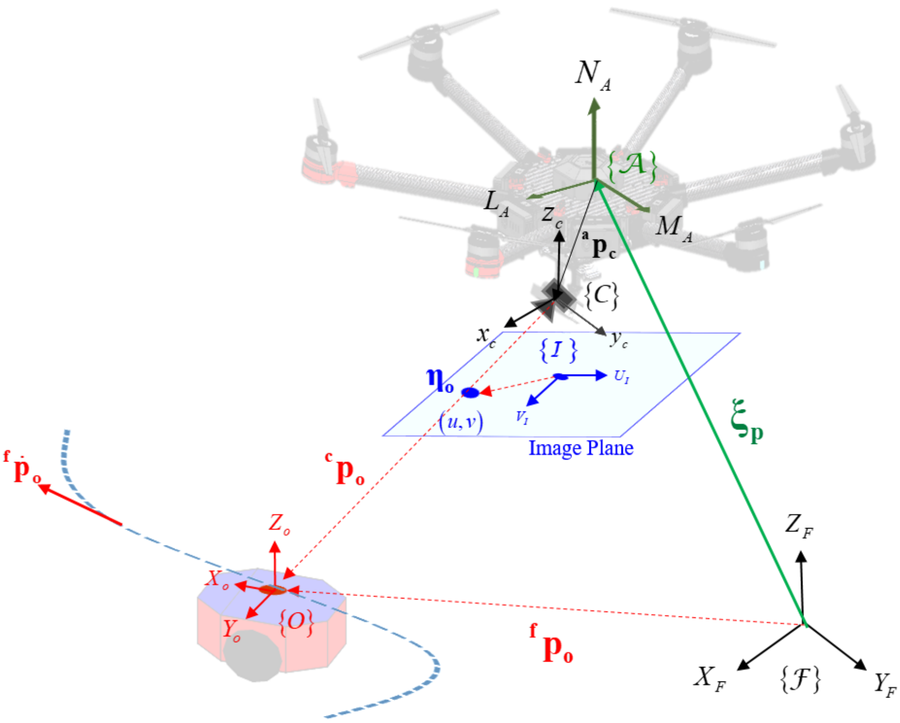


Figure 2. Schematic diagram of camera configuration on the UAV. The target of interest to be followed is represented by a unicycle-type mobile robot and an aerial robot with a camera located at the bottom.

3.2. The IBVS Model of the UAV

Now, the behavior of the object as a function of the robot's velocities is detailed. For this purpose, a pinhole camera model is considered [25], where the position of the object of interest ${}^{\mathbf{c}}\mathbf{p_o} \in \Re^3$ is projected onto the image plane. This relationship between the camera reference system and the image plane is given by

$$\boldsymbol{\eta}({}^{c}x_o, {}^{c}y_o, {}^{c}z_o) = \begin{bmatrix} u \\ v \end{bmatrix} = \frac{f}{{}^{c}z_o} \begin{bmatrix} {}^{c}x_o \\ {}^{c}y_o \end{bmatrix} \tag{11}$$

where $\boldsymbol{\eta}({}^{c}x_o, {}^{c}y_o, {}^{c}z_o)$ is the feature vector of the target on the image plane and $f \in \Re$ is the focal length of the camera. The time derivative of (11) is given by

$$\dot{\boldsymbol{\eta}} = -f \begin{bmatrix} \frac{1}{{}^{c}z_o} & 0 & -\frac{{}^{c}x_o}{{}^{c}z_o^2} \\ 0 & \frac{1}{{}^{c}z_o} & -\frac{{}^{c}y_o}{{}^{c}z_o^2} \end{bmatrix} \begin{bmatrix} {}^{c}\dot{x}_o \\ {}^{c}\dot{y}_o \\ {}^{c}\dot{z}_o \end{bmatrix} \tag{12}$$

Now, let ${}^{\mathbf{f}}\mathbf{p_o}(t) \in \Re^3$ be the position of the object of interest in the fixed reference frame $\{\mathcal{F}\}$ and ${}^{\mathbf{f}}\mathbf{p_c}(t) \in \Re^3$ be the camera position in $\{\mathcal{F}\}$. Then, the position of the object of interest with respect to the frame of the camera ${}^{\mathbf{c}}\mathbf{p_t}(t) \in \Re^3$ is given by

$${}^{\mathbf{c}}\mathbf{p_o}(t) = {}^{\mathbf{c}}\mathbf{R_f}\left({}^{\mathbf{f}}\mathbf{p_o}(t) - {}^{\mathbf{f}}\mathbf{p_c}(t)\right) \tag{13}$$

where ${}^{\mathbf{c}}\mathbf{p_o}(t) = \begin{bmatrix} {}^{c}x_o & {}^{c}y_o & {}^{c}z_o \end{bmatrix}^T$ is the position of the object of interest in the camera's reference system $\{\mathcal{C}\}$ and ${}^{\mathbf{c}}\mathbf{R_f} = \left({}^{\mathbf{f}}\mathbf{R_c}\right)^T = \left({}^{\mathbf{f}}\mathbf{R_c}\right)^{-1}$ is the inverse matrix of the camera rotation with respect to the fixed system $\{\mathcal{F}\}$. Now, if the camera is considered as a moving body with linear and angular velocity denoted by ${}^{\mathbf{f}}\mathbf{v_c}(t) = \begin{bmatrix} v_{x_c} & v_{y_c} & v_{z_c} \end{bmatrix}^T \in \Re^3$ and

${}^{\mathbf{f}}\boldsymbol{\omega}_{\mathbf{c}}(t) = \begin{bmatrix} \omega_{x_c} & \omega_{y_c} & \omega_{z_c} \end{bmatrix}^T \in \Re^3$, respectively, then the target velocity is obtained from the time derivative of (13), such that

$$ {}^{\mathbf{c}}\dot{\mathbf{p}}_{\mathbf{o}} = {}^{\mathbf{c}}\mathbf{R}_{\mathbf{f}}\left({}^{\mathbf{f}}\dot{\mathbf{p}}_{\mathbf{o}} - {}^{\mathbf{f}}\mathbf{v}_{\mathbf{c}}\right) - {}^{\mathbf{c}}\mathbf{R}_{\mathbf{f}}\left[{}^{\mathbf{f}}\boldsymbol{\omega}_{\mathbf{c}} \times \left({}^{\mathbf{f}}\mathbf{p}_{\mathbf{o}} - {}^{\mathbf{f}}\mathbf{p}_{\mathbf{c}}\right)\right] \tag{14} $$

By applying the corresponding operations and the properties of the cross-product, we obtain that the velocity of a point in space as a function of the velocities of the camera, which is given by

$$ \begin{bmatrix} {}^{c}\dot{x}_o \\ {}^{c}\dot{y}_o \\ {}^{c}\dot{z}_o \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 & 0 & -{}^{c}z_o & {}^{c}y_o \\ 0 & -1 & 0 & {}^{c}z_o & 0 & -{}^{c}x_o \\ 0 & 0 & -1 & -{}^{c}y_o & {}^{c}x_o & 0 \end{bmatrix} \begin{bmatrix} {}^{\mathbf{c}}\mathbf{R}_{\mathbf{f}}{}^{\mathbf{f}}\mathbf{v}_{\mathbf{c}} \\ {}^{\mathbf{c}}\mathbf{R}_{\mathbf{f}}{}^{\mathbf{f}}\boldsymbol{\omega}_{\mathbf{c}} \end{bmatrix} + {}^{\mathbf{c}}\mathbf{R}_{\mathbf{f}}{}^{\mathbf{f}}\dot{\mathbf{p}}_{\mathbf{o}} \tag{15} $$

where ${}^{\mathbf{f}}\dot{\mathbf{p}}_{\mathbf{o}} \in \Re^3$ is the velocity of the moving target; in the case where the object is static, ${}^{\mathbf{f}}\dot{\mathbf{p}}_{\mathbf{o}} = \mathbf{0}$. Then, substituting (15) into (12) results in

$$ \dot{\boldsymbol{\eta}} = -f \begin{bmatrix} \frac{1}{{}^{c}z_o} & 0 & -\frac{{}^{c}x_o}{{}^{c}z_o^2} \\ 0 & \frac{1}{{}^{c}z_o} & -\frac{{}^{c}y_o}{{}^{c}z_o^2} \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 & 0 & -{}^{c}z_o & {}^{c}y_o \\ 0 & -1 & 0 & {}^{c}z_o & 0 & {}^{c}x_o \\ 0 & 0 & -1 & -{}^{c}y_o & {}^{c}x_o & 0 \end{bmatrix} \begin{bmatrix} {}^{\mathbf{c}}\mathbf{R}_{\mathbf{f}}{}^{\mathbf{f}}\mathbf{v}_{\mathbf{c}} \\ {}^{\mathbf{c}}\mathbf{R}_{\mathbf{f}}{}^{\mathbf{f}}\boldsymbol{\omega}_{\mathbf{c}} \end{bmatrix} - f \begin{bmatrix} \frac{1}{{}^{c}z_o} & 0 & -\frac{{}^{c}x_o}{{}^{c}z_o^2} \\ 0 & \frac{1}{{}^{c}z_o} & -\frac{{}^{c}y_o}{{}^{c}z_o^2} \end{bmatrix} {}^{\mathbf{c}}\mathbf{R}_{\mathbf{f}}{}^{\mathbf{f}}\dot{\mathbf{p}}_{\mathbf{o}} \tag{16} $$

which, written in a compact form, is

$$ \dot{\boldsymbol{\eta}}(t) = \mathbf{J}_{\mathbf{I}}(\boldsymbol{\eta}, {}^{\mathbf{c}}\mathbf{p}_{\mathbf{o}}) \begin{bmatrix} {}^{\mathbf{c}}\mathbf{R}_{\mathbf{f}}{}^{\mathbf{f}}\mathbf{v}_{\mathbf{c}}(t) \\ {}^{\mathbf{c}}\mathbf{R}_{\mathbf{f}}{}^{\mathbf{f}}\boldsymbol{\omega}_{\mathbf{c}}(t) \end{bmatrix} - f \begin{bmatrix} \frac{1}{{}^{c}z_o} & 0 & -\frac{{}^{c}x_o}{{}^{c}z_o^2} \\ 0 & \frac{1}{{}^{c}z_o} & -\frac{{}^{c}y_o}{{}^{c}z_o^2} \end{bmatrix} {}^{\mathbf{c}}\mathbf{R}_{\mathbf{f}}{}^{\mathbf{f}}\dot{\mathbf{p}}_{\mathbf{o}}(t) \tag{17} $$

Finally, as the camera is mounted on the UAV, let us express (17) as a function of the UAV velocities as follows:

$$ \dot{\boldsymbol{\eta}}(t) = \mathbf{J}(\boldsymbol{\eta}, {}^{\mathbf{c}}\mathbf{p}_{\mathbf{o}}, \mathbf{q})\mathbf{u}(t) + \mathbf{J}_{\mathbf{o}}(\mathbf{q}, {}^{\mathbf{c}}\mathbf{p}_{\mathbf{o}}){}^{\mathbf{f}}\dot{\mathbf{p}}_{\mathbf{o}}(t) \tag{18} $$

with

$$ {}^{\mathbf{c}}\mathbf{J}(\boldsymbol{\eta}, {}^{\mathbf{c}}\mathbf{p}_{\mathbf{o}}, \mathbf{q}) = \mathbf{J}_{\mathbf{I}}(\boldsymbol{\eta}, {}^{\mathbf{c}}\mathbf{p}_{\mathbf{o}}) \begin{bmatrix} {}^{\mathbf{c}}\mathbf{R}_{\mathbf{f}} & \mathbf{0}_{3\times 3} \\ \mathbf{0}_{3\times 3} & {}^{\mathbf{c}}\mathbf{R}_{\mathbf{f}} \end{bmatrix} \begin{bmatrix} \mathbf{I}_{\mathbf{3}\times\mathbf{3}} & \mathbf{0}_{\mathbf{3}\times\mathbf{3}} \\ \mathbf{0}_{\mathbf{3}\times\mathbf{3}} & \mathbf{T}_{\mathbf{A}}(\mathbf{q}) \end{bmatrix} \mathbf{J}_{\mathbf{A}}(\mathbf{q}) \tag{19} $$

and

$$ \mathbf{J}_{\mathbf{o}}(\mathbf{q}, {}^{\mathbf{c}}\mathbf{p}_{\mathbf{o}}) = -f \begin{bmatrix} \frac{1}{{}^{c}z_o} & 0 & -\frac{{}^{c}x_o}{{}^{c}z_o^2} \\ 0 & \frac{1}{{}^{c}z_o} & -\frac{{}^{c}y_o}{{}^{c}z_o^2} \end{bmatrix} {}^{\mathbf{c}}\mathbf{R}_{\mathbf{f}} \tag{20} $$

where $\mathbf{T}_{\mathbf{A}}(\mathbf{q})$ is a transformation matrix that depends on the parameterization of the UAV operating point orientation.

$$ \dot{\boldsymbol{\eta}}(t) = \mathbf{J}\mathbf{u}(t) + \dot{\boldsymbol{\eta}}_{\mathbf{o}}(t) \tag{21} $$

where $\dot{\boldsymbol{\eta}}_{\mathbf{o}}(t) = \mathbf{J}_{\mathbf{o}}(\mathbf{q}, {}^{\mathbf{c}}\mathbf{p}_{\mathbf{o}}){}^{\mathbf{f}}\dot{\mathbf{p}}_{\mathbf{o}}(t)$ is the velocity of the object, which, for short, can be written as $\dot{\boldsymbol{\eta}}_{\mathbf{o}}(t) = \mathbf{J}_{\mathbf{o}}\mathbf{u}_{\mathbf{o}}(\mathbf{t})$.

4. Controller

Figure 3 shows the scheme of the controller proposed in this work for tracking targets.

The control scheme is composed of two stages. *(i) Visual servo control based on the sliding-mode technique*: this stage is responsible for generating the maneuverability velocities of the UAV to follow the target of interest. The controller is designed under the sliding-mode technique. The references received by the controller are the desired position of the target in the image plane and the estimated velocity of the target.

In addition, the scheme can add secondary objectives to the controller without modifying the primary objective. These secondary tasks can be obstacle avoidance, desired tracking orientation, and the limitation of robot maneuverability actions, among others. In this work, we consider as the secondary task the tracking angle, which decides which angle the UAV uses to follow the target of interest; *(ii) Controller based on the UAV dynamics*: this cascade controller allows one to compensate the control actions, i.e., the controller guarantees that the robot follows the actions generated by the primary controller. This controller is known as a dynamics compensator; as its name implies, it compensates for dynamics not considered in the robot's model and/or disturbances generated at the time of performing the task. The controller receives as reference the velocities of the servo visual controller and the actual velocities of the UAV; with these two references, a velocity error is obtained. Therefore, the objective of this controller is to drive the velocity error to zero.

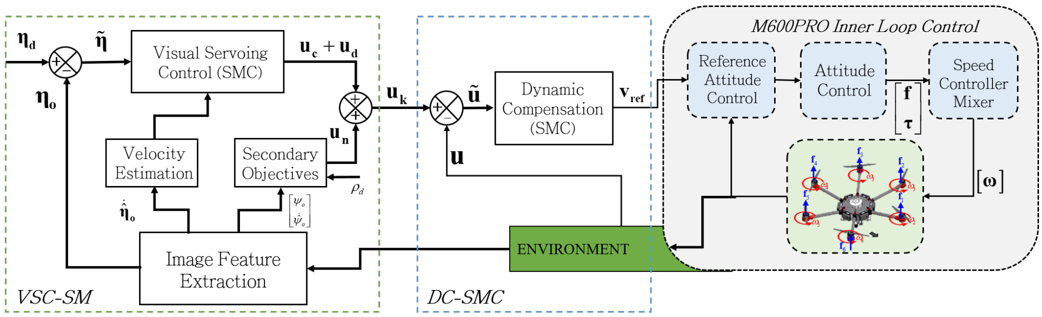


Figure 3. Proposed cascade servo visual control scheme for target tracking based on sliding mode.

4.1. Design of the Visual Servo Controller Based on Sliding Mode (VSC-SM)

Let $\tilde{\boldsymbol{\eta}}(t) = \boldsymbol{\eta}_{\mathbf{d}} - \boldsymbol{\eta}(t)$ be the tracking error of the target and $\dot{\tilde{\boldsymbol{\eta}}}(t) = -\dot{\boldsymbol{\eta}}(t)$ be the error velocity. A time-varying sliding surface s in the state space $\Re(n)$ is proposed by the following scalar equation [26]:

$$s(t,\tilde{\boldsymbol{\eta}}) = \left(\frac{d}{dt} + \mathbf{K_n}\right)^{n-1} \int_0^t \tilde{\boldsymbol{\eta}}(\tau)d\tau \tag{22}$$

where $\mathbf{K_n} \in \Re^{2\times 2}$ is a positive diagonal matrix. Taking the number of states $n = 2$, Equation (22) results in

$$s(t,\tilde{\boldsymbol{\eta}}) = \tilde{\boldsymbol{\eta}}(t) + \mathbf{K_n} \int_0^t \tilde{\boldsymbol{\eta}}(\tau)d\tau \tag{23}$$

Now, let us see the dynamic behavior of the system (23) using the time derivative:

$$\dot{s}(t,\tilde{\boldsymbol{\eta}}) = \dot{\tilde{\boldsymbol{\eta}}}(t) + \mathbf{K_n}\tilde{\boldsymbol{\eta}}(t) \tag{24}$$

considering that we are on the sliding surface, i.e., we have zero tracking error. If $\dot{s} = \mathbf{0}$, then $\dot{s}(t,\tilde{\boldsymbol{\eta}}) = \dot{\tilde{\boldsymbol{\eta}}}(t) + \mathbf{K_n}\tilde{\boldsymbol{\eta}}(t) = \mathbf{0}$; substituting the derivative of $\tilde{\boldsymbol{\eta}}(t)$ results in

$$\dot{s}(t,\tilde{\boldsymbol{\eta}}) = -\dot{\boldsymbol{\eta}}(t) + \mathbf{K_n}\tilde{\boldsymbol{\eta}}(t) = \mathbf{0} \tag{25}$$

replacing the system model in (21) with (26), we obtain

$$\dot{s}(t,\tilde{\boldsymbol{\eta}}) = -\mathbf{J}\mathbf{u}(t) - \dot{\boldsymbol{\eta}}_{\mathbf{o}}(t) + \mathbf{K_n}\tilde{\boldsymbol{\eta}}(t) = \mathbf{0} \tag{26}$$

Thus, the best approximation of the continuous control action is proposed as $\mathbf{u_c}(t)$, which allows $\dot{s} = 0$. If $\mathbf{J}\mathbf{u_c}(t) = -\dot{\hat{\boldsymbol{\eta}}}_{\mathbf{o}}(t) + \mathbf{K_n}\tilde{\boldsymbol{\eta}}(t)$, the action, therefore, results in

$$\mathbf{u_c}(t) = \mathbf{J}^{\dagger}\left(-\dot{\hat{\boldsymbol{\eta}}}_{\mathbf{o}}(t) + \mathbf{K_n}\tilde{\boldsymbol{\eta}}(t)\right) \tag{27}$$

where $\dot{\hat{\boldsymbol{\eta}}}_{\mathbf{o}}(t)$ is the estimated velocity of the object in the image plane $\{\mathcal{I}\}$, which is given by $\dot{\hat{\boldsymbol{\eta}}}_{\mathbf{o}}(t) = \mathbf{J_o}\dot{\hat{\mathbf{u}}}_{\mathbf{o}}$ ($\dot{\hat{\mathbf{u}}}_{\mathbf{o}}$, which is the estimated velocity of the object with respect to $\{\mathcal{F}\}$), and $\mathbf{J}^{\dagger}$ is the Moore–Penrose pseudoinverse [27] of $\mathbf{J}$, which is defined by $\mathbf{J}^{\dagger} = \mathbf{W}^{-1}\mathbf{J}^{T}\left(\mathbf{J}\mathbf{W}^{-1}\mathbf{J}^{T}\right)^{-1}$, where $\mathbf{W} \in \Re^{n\times n}$ is a positive definite square matrix that weighs the minimum norm solution of $\mathbf{J}^{\dagger}$. By means of this matrix, it is possible to decide which solutions to the system in (26) are relevant; else, if $\mathbf{W} = \mathbf{I}_{n\times n}$, then $\mathbf{J}^{\dagger} = \mathbf{J}^{T}\left(\mathbf{J}\mathbf{J}^{T}\right)^{-1}$ (the pseudo-inverse) is equally weighted for each control action.

Now, in order to accomplish the sliding condition $\frac{1}{2}\frac{d}{dt}s^2 \leq -\lambda(\mathbf{K_d})\|s\|$ [26,28], to the previously obtained controlling control action in (27), we add a discontinuous control action defined as $\mathbf{u_d}(t)$. Therefore, the complete control action is defined by

$$\mathbf{u_k}(t) = \mathbf{u_c}(t) + \mathbf{u_d}(t) \tag{28}$$

Let us see how the control action $\mathbf{u_d}(t)$ is given as a function of the stability of the system. We define a Lyapunov candidate $V = \frac{1}{2}s^T s$, with time derivative $\dot{V} = s^T\dot{s}$. Replacing $\dot{s}$ from (26), we obtain

$$\dot{V} = s^T(-\mathbf{J}\mathbf{u} - \dot{\boldsymbol{\eta}}_{\mathbf{o}} + \mathbf{K_n}\tilde{\boldsymbol{\eta}}) \tag{29}$$

Applying the control action, we have that $\mathbf{u}(t) \approx \mathbf{u_k}(t)$, i.e., that the system is on the sliding surface. As a result, we obtain

$$\begin{aligned}
\dot{V} &= s^T(-\mathbf{J}(\mathbf{u_c} + \mathbf{u_d}+) - \dot{\boldsymbol{\eta}}_{\mathbf{o}} + \mathbf{K_n}\tilde{\boldsymbol{\eta}}) \\
&= s^T(-\mathbf{J}\mathbf{u_c} - \mathbf{J}\mathbf{u_d} - \dot{\boldsymbol{\eta}}_{\mathbf{o}} + \mathbf{K_n}\tilde{\boldsymbol{\eta}}) \\
&= s^T\left(-\mathbf{J}\left(\mathbf{J}^{\dagger}(-\dot{\hat{\boldsymbol{\eta}}}_{\mathbf{o}} + \mathbf{K_n}\tilde{\boldsymbol{\eta}})\right) - \mathbf{J}\mathbf{u_d} - \dot{\boldsymbol{\eta}}_{\mathbf{o}} + \mathbf{K_n}\tilde{\boldsymbol{\eta}}\right) \\
&= s^T\left(\mathbf{J}\mathbf{J}^{\dagger}\dot{\hat{\boldsymbol{\eta}}}_{\mathbf{o}} - \mathbf{J}\mathbf{J}^{\dagger}\mathbf{K_n}\tilde{\boldsymbol{\eta}} - \mathbf{J}\mathbf{u_d} - \dot{\boldsymbol{\eta}}_{\mathbf{o}} + \mathbf{K_n}\tilde{\boldsymbol{\eta}}\right)
\end{aligned} \tag{30}$$

Now, it is assumed that the estimated velocity of the object is equal to the actual velocity $\mathbf{J_o}\hat{\mathbf{u}}_{\mathbf{o}} \equiv \mathbf{J_o}\mathbf{u_o}$; hence, $\dot{\hat{\boldsymbol{\eta}}}_{\mathbf{o}} = \dot{\boldsymbol{\eta}}_{\mathbf{o}}$. The derivative of the Lyapunov candidate results in

$$\dot{V} = s^T\left(\left(\mathbf{J}\mathbf{J}^{\dagger} - \mathbf{I}\right)\dot{\boldsymbol{\eta}}_{\mathbf{o}} - \left(\mathbf{J}\mathbf{J}^{\dagger} - \mathbf{I}\right)\mathbf{K_n}\tilde{\boldsymbol{\eta}} - \mathbf{J}\mathbf{u_d}\right) \tag{31}$$

Finally, as $\mathbf{J}\mathbf{J}^{\dagger} = \mathbf{I} \to \left(\mathbf{J}\mathbf{J}^{\dagger} - \mathbf{I}\right) = \mathbf{0}$, we obtain that

$$\dot{V} = s^T(-\mathbf{J}\mathbf{u_d}) \tag{32}$$

For the system to be asymptotically stable, $\dot{V} < 0$ must be satisfied. Therefore, from (32), we propose discontinuous action as

$$\mathbf{u_d}(s) = \mathbf{J}^{\dagger}\mathbf{K_d}f(s) \tag{33}$$

where $f(s) = sat(s,\varepsilon)$, which is a nonlinear function, saturated in such a way as to reduce the effect of chattering (Chattering Using Quasi-Sliding-Mode Control) [29]. And this is defined as

$$sat(s_i,\varepsilon) = \begin{cases} 1 & \text{if} \quad s_i > \varepsilon \\ \frac{s_i}{\varepsilon} & \text{if} \quad -\varepsilon \leq s_i \leq \varepsilon \\ -1 & \text{if} \quad s_i < -\varepsilon \end{cases} \tag{34}$$

where $\varepsilon \in \Re^{+}$ is the boundary layer thickness of the chattering. This saturation function in (34) scales smoothly between -1 and 1 when s is within the interval of $[-\varepsilon,\varepsilon]$, and $\mathbf{K_d} \in \Re^{2\times 2}$ is a positive definite matrix weighing the discontinuous action. Therefore, the control action is defined as

$$\mathbf{u_d}(s) = \mathbf{J}^{\dagger}\mathbf{K_d}sat(s,\varepsilon) \tag{35}$$

Therefore, if we replace (32) in the derivative of the Lyapunov candidate, we obtain that

$$\begin{aligned}\dot{V} &= s^T\left(-\mathbf{J}\mathbf{J}^{\dagger}\mathbf{K_d}sat(s,\varepsilon)\right)\\ &= -s^T(\mathbf{K_d}sat(s,\varepsilon))\\ &\leq -\lambda_{min}(\mathbf{K_d})\|s\|\end{aligned} \tag{36}$$

Consequently, the controller satisfies the sliding condition in the SMC design given by $\frac{1}{2}\frac{d}{dt}s^2 \leq -\lambda_{min}(\mathbf{K_d})\|s\|$. This condition establishes that the square to the sliding surface must be close to zero, i.e., the distance to the surface decreases along all trajectories of the system. Therefore, by this condition, it is intended that the states of the system approach the sliding surface and, once on the surface, they stay there. In other words, the sliding condition makes the sliding surface an invariant set.

Secondary Objectives

Now, as can be seen, the control action is a control action determined by a minimum norm solution to the system in (26), which is a redundant system; therefore, the homogeneous solution can be obtained, which, in this case, represents a control action in the null space of $\mathbf{J}$. This control action generates internal robot configuration motions without affecting the velocity of the operating point, i.e., it does not affect the main task; therefore, secondary control objectives can be defined as $\mathbf{u_n}(t) = (\mathbf{I} - \mathbf{J}^{\dagger}\mathbf{J})\mathbf{\Delta_u}(t)$. Consequently, the total controller is given by

$$\mathbf{u_k}(t) = \mathbf{u_c}(t) + \mathbf{u_d}(t) + \mathbf{u_n}(t) \tag{37}$$

By substituting the corresponding control actions (27), (33), and (37), the proposed servo visual controller is defined as follows:

$$\mathbf{u_k}(t) = \mathbf{J}^{\dagger}\left(-\dot{\boldsymbol{\eta}}_{\mathbf{o}}(t) + \mathbf{K_n}\tilde{\boldsymbol{\eta}}(t) + \mathbf{K_d}sat(s,\varepsilon)\right) + \left(\mathbf{I} - \mathbf{J}^{\dagger}\mathbf{J}\right)\mathbf{\Delta_u}(t) \tag{38}$$

where $\mathbf{\Delta_u} \in \mathbb{R}^n$ is an arbitrary velocity projected into the null space of $\mathbf{J}$. This arbitrary velocity defines the secondary objectives, such as obstacle avoidance, movement constraints, and robot orientation, among others. For this case, we consider two secondary objectives:

- Safe Tracking Distance: This objective is established given that the system model in Equation (21) focuses on a point in the image plane, so the UAV can position itself in any way relative to the target. Consequently, to guarantee a safe position of the UAV in relation to the target, it is proposed to maintain a safe tracking distance;
- UAV Orientation Relative to the Target: This second objective defines the correct orientation of the UAV based on the target's position.

Now, let ${}^{\mathrm{f}}\mathbf{P_o} = \begin{bmatrix}{}^{\mathrm{f}}x_o & {}^{\mathrm{f}}y_o & {}^{\mathrm{f}}z_o\end{bmatrix}^T$ be the position of the object in $\{\mathcal{F}\}$'s system reference and $\boldsymbol{\xi}_{\mathbf{p}}$ be the position of the UAV in $\{\mathcal{F}\}$'s system reference. The distance of the UAV relative to the target is defined by

$$\rho = \sqrt{({}^{\mathrm{f}}x_o - \xi_x)^2 + ({}^{\mathrm{f}}y_o - \xi_y)^2 + ({}^{\mathrm{f}}z_o - \xi_z)^2)} \tag{39}$$

with time derivative

$$\dot{\rho}(t) = \begin{bmatrix}\frac{\xi_x - {}^{\mathrm{f}}x_o}{\rho} & \frac{\xi_y - {}^{\mathrm{f}}y_o}{\rho} & \frac{\xi_z - {}^{\mathrm{f}}z_o}{\rho}\end{bmatrix}\begin{bmatrix}\dot{\xi}_x(t)\\ \dot{\xi}_y(t)\\ \dot{\xi}_z(t)\end{bmatrix} \tag{40}$$

Finally, a Jacobian is proposed to control both the distance ρ and the UAV orientation $\xi_\psi(t)$ as follows:

$$\begin{bmatrix} \dot{\rho}(t) \\ \dot{\xi}_\psi(t) \end{bmatrix} = \begin{bmatrix} \frac{\xi_x - {}^f x_o}{\rho} & \frac{\xi_y - {}^f y_o}{\rho} & \frac{\xi_z - {}^f z_o}{\rho} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{\xi}_x(t) \\ \dot{\xi}_y(t) \\ \dot{\xi}_z(t) \\ \dot{\xi}_\psi(t) \end{bmatrix} \tag{41}$$

Now, for the purpose of representing System (41) in terms of the robot's maneuverability velocities, we use the Jacobian of the robot described in (4). Therefore, the system can be expressed as

$$\begin{bmatrix} \dot{\rho}(t) \\ \dot{\xi}_\psi(t) \end{bmatrix} = \begin{bmatrix} \frac{\xi_x - {}^f x_o}{\rho} & \frac{\xi_y - {}^f y_o}{\rho} & \frac{\xi_z - {}^f z_o}{\rho} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(\psi) & -\sin(\psi) & 0 & 0 \\ \sin(\psi) & \cos(\psi) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u_l \\ u_m \\ u_n \\ u_\psi \end{bmatrix} \tag{42}$$

Written in matrix form, the differential kinematics of the proposed secondary task (secondary objectives) is given by

$$\boldsymbol{\xi}^*(t) = \mathbf{J_n}(\rho, \psi)\mathbf{u}(t) \tag{43}$$

where $\boldsymbol{\xi}^*(t) = \begin{bmatrix} \dot{\rho}(t) & \dot{\xi}_\psi(t) \end{bmatrix}^\intercal$ is the velocity vector of the proposed secondary objectives and $\mathbf{J_n} \in \mathbb{R}^{2\times 4}$ is the Jacobian that maps the velocities of the secondary task to the maneuverability velocities of the UAV. Now, to obtain the velocity of the UAV to be projected in the null space $\boldsymbol{\Delta}_u$, an inverse kinematic control is used. The control of the secondary task is given by

$$\boldsymbol{\Delta}_\mathbf{u}(t) = \mathbf{J_n}^\dagger \left(\dot{\boldsymbol{\xi}}_d^*(t) + \mathbf{K_o} \tanh\left(\tilde{\boldsymbol{\xi}}_d^*(t)\right) \right) \tag{44}$$

where $\mathbf{J_n}^\dagger$ is the pseudo inverse matrix of the Jacobian of the secondary task; $\dot{\boldsymbol{\xi}}_d^*(t) = \begin{bmatrix} 0 & \dot{\hat{\psi}}_o(t) \end{bmatrix}^\intercal$ is the desired velocity of the task, where $\dot{\hat{\psi}}_o$ is the estimated velocity of the target; $\tilde{\boldsymbol{\xi}}_d^*(t) = \begin{bmatrix} \tilde{\rho}(t) & \tilde{\xi}_\psi(t) \end{bmatrix}^\intercal$ is the secondary task error vector, where $\tilde{\rho}(t) = \rho_d - \rho(t)$ is the safe distance error obtained by subtracting from the desired distance ρ_d and the distance of the UAV from the target ρ; y $\tilde{\xi}_\psi(t) = \psi_o(t) - \xi_\psi(t)$ is the orientation error, where $\psi_o(t)$ is the target orientation; $\mathbf{K_o} = diag\left(\begin{bmatrix} k_\rho & k_o \end{bmatrix}\right)$ is a positive definite diagonal matrix that weighs the control errors of the secondary task with $k_\rho, k_o > 0$; and $\tanh(\bullet)$ is an odd function that saturates the orientation and safe distance error, such that

$$\boldsymbol{\Delta}_\mathbf{u}(t) = \mathbf{J_n}^\dagger \begin{bmatrix} k_\rho \tanh(\rho_d - \rho) \\ \dot{\hat{\psi}}_o + k_o \tanh(\psi_o - \psi) \end{bmatrix} \tag{45}$$

For the distance error, a variable gain is proposed, depending on the error norm of the main task $k_\rho = f(\sigma, \|\tilde{\boldsymbol{\eta}}\|)$, which is a positive constant that weights the orientation of the UAV with respect to the target. It is defined as a function of the tracking error norm and a positive weighting constant and is given as

$$k_\rho = \frac{\sigma}{1 + \|\tilde{\boldsymbol{\eta}}\|} \tag{46}$$

where $\sigma \in \Re^+$ represents the maximum saturation value of k_ρ. Finally, considering (38) and (44), it turns out that the proposed control for tracking a moving target based on sliding mode results in

$$\mathbf{u_k}(t) = \mathbf{J}^\dagger\left(-\dot{\boldsymbol{\eta}}_\mathbf{o}(t) + \mathbf{K_n}\tilde{\boldsymbol{\eta}}(t) + \mathbf{K_d} sat(s, \varepsilon)\right) + \left(\mathbf{I} - \mathbf{J}^\dagger \mathbf{J}\right)\mathbf{J_n}^\dagger \left(\dot{\boldsymbol{\xi}}_d^*(t) + \mathbf{K_o} \tanh\left(\tilde{\boldsymbol{\xi}}_d^*(t)\right)\right) \tag{47}$$

4.2. Dynamic Compensation Sliding-Mode Control (DC-SMC)

In this section, the dynamic compensator is presented. The purpose is to achieve a perfect tracking of the reference velocity. Considering the inverse dynamic model of the

UAV in (10), the direct dynamic model is obtained by finding the acceleration $\dot{\mathbf{u}}(t)$, such that $\dot{\mathbf{u}}(t) = \mathbf{M}^{-1}(\delta)(\mathbf{u_{ref}}(t) - \mathbf{C}(\delta)\mathbf{u}(t) - \zeta(t))$, $\tilde{\mathbf{u}}(t) = \mathbf{u_k}(t) - \mathbf{u}(t)$ is the velocity error, and $\dot{\tilde{\mathbf{u}}}(t) = \dot{\mathbf{u}}_\mathbf{k}(t) - \dot{\mathbf{u}}(t)$ is the derivative of the velocity error. For its development in the same way as in (22), a sliding surface is proposed as a function of the velocity error $s(t, \tilde{\mathbf{u}})$ as

$$s(t, \tilde{\mathbf{u}}) = \tilde{\mathbf{u}}(t) + \mathbf{K_u} \int_0^t \tilde{\mathbf{u}}(\tau) d\tau \tag{48}$$

with $\mathbf{K_u} \in \Re^{4\times 4}$ as a positive diagonal matrix that weights the velocity error and is given by $\mathbf{K_u} = diag(k_{u1}, k_{u2}, k_{u3}, k_{u4})$, with $ku_i > 0$ for $i = 1, ..., 4$. Now, the time derivative of the sliding surface $s(t, \tilde{\mathbf{u}})$ is

$$\dot{s}(t, \tilde{\mathbf{u}}) = \dot{\tilde{\mathbf{u}}}(t) + \mathbf{K_u}\tilde{\mathbf{u}}(t) \tag{49}$$

Then, the control action is given such that the sliding surface is reached in a finite time t_a. The controller is obtained from the sliding condition:

$$\tfrac{1}{2}\tfrac{d}{dt}s^T s \leq -k_\alpha \|s\| \quad \forall t \geq 0,\ k_\alpha > 0 \tag{50}$$

In order to obtain the control action that determines the dynamics of the system on the sliding surface, we take into account the condition of the stability of control in sliding mode $\dot{s}(t, \tilde{\mathbf{u}}) = \mathbf{0}$; therefore, by replacement in (48),

$$\begin{aligned} \dot{s} &= \dot{\tilde{\mathbf{u}}} + \mathbf{K_u}\tilde{\mathbf{u}} = 0 \\ &= \dot{\mathbf{u}}_\mathbf{k} - \dot{\mathbf{u}} + \mathbf{K_u}\tilde{\mathbf{u}} = 0 \\ &= \dot{\mathbf{u}}_\mathbf{k} - \mathbf{M}^{-1}(\mathbf{u_{ref}} - \mathbf{Cu} - \zeta) + \mathbf{K_u}\tilde{\mathbf{u}} = 0 \end{aligned} \tag{51}$$

As a control action, $\mathbf{v_{ref}}$ is required to satisfy the condition of the dynamic model equation with (51). Consequently, the following controller is proposed:

$$\mathbf{v_{ref}}(t) = \mathbf{M}(\delta)(\dot{\mathbf{u}}_\mathbf{k}(t) + \mathbf{K_u}\tilde{\mathbf{u}}(t)) + \mathbf{C}(\delta)\mathbf{u}(t) + \mathbf{K_s} sign(s) \tag{52}$$

where $sign(s)$ is a discontinuous function given by

$$sign(s_i) = \begin{cases} 1 & \text{if} \quad s_i > 0 \\ 0 & \text{if} \quad s_i = 0 \\ -1 & \text{if} \quad s_i < 0 \end{cases} \tag{53}$$

where $\mathbf{K_s} = diag(k_{s1}, ..., k_{s4})$ is the gain matrix of the SMC condition. Moreover, the vector of disturbances and/or errors in the model (10) results in $\|\zeta\| \leq b$; then, $b \leq \lambda_{\min}(\mathbf{K_s}) \to \|\zeta\| \leq \lambda_{\min}(\mathbf{K_s})$. Taking these considerations into account, we define the following Lyapunov candidate function $V(s) = \frac{1}{2}\frac{d}{dt}s^T s$: the time derivative of the proposed candidate is $\dot{V}(s) = s^T\dot{s}$. Now, substituting (51) and applying the control action into the system ($\mathbf{u_{ref}} \equiv \mathbf{v_{ref}}$) results in

$$\begin{aligned} \dot{V} &= s^T\left[\dot{\mathbf{u}}_\mathbf{k} - \mathbf{M}^{-1}(\mathbf{u_{ref}} - \mathbf{Cu} - \zeta) + \mathbf{K_u}\tilde{\mathbf{u}}\right] \\ &= s^T\left[\dot{\mathbf{u}}_\mathbf{k} - \mathbf{M}^{-1}[\mathbf{M}(\dot{\mathbf{u}}_\mathbf{k} + \mathbf{K_u}\tilde{\mathbf{u}}) + \mathbf{Cu} + \mathbf{K_s} sign(s) - \mathbf{Cu} - \zeta] + \mathbf{K_u}\tilde{\mathbf{u}}\right] \\ &= s^T\left[\dot{\mathbf{u}}_\mathbf{k} - \mathbf{M}^{-1}\left[\mathbf{M}\dot{\mathbf{u}}_\mathbf{k} + \mathbf{M}\mathbf{K_u}\dot{\tilde{\mathbf{u}}} + \mathbf{K_s} sign(s) - \zeta\right] + \mathbf{K_u}\tilde{\mathbf{u}}\right] \\ &= s^T\left[-\mathbf{M}^{-1}\mathbf{K_s} sign(s) + \mathbf{M}^{-1}\zeta\right] = -\left\|s^T\mathbf{M}^{-1}\mathbf{K_s} sign(s)\right\| + \left\|s^T\mathbf{M}^{-1}\zeta\right\| \\ &\leq -\left\|\mathbf{M}^{-1}\mathbf{K_s}\right\|\|s\| + \left\|\mathbf{M}^{-1}\zeta\right\|\|s\| \\ &\leq -\lambda_{min}(\mathbf{M}^{-1})\lambda_{min}(\mathbf{K_s})\|s\| + \lambda_{min}(\mathbf{M}^{-1})\|\zeta\|\|s\| \\ \dot{V} &\leq -\lambda_{min}(\mathbf{M}^{-1})(\lambda_{min}(\mathbf{K_s}) - \|\zeta\|)\|s\| \end{aligned} \tag{54}$$

As mentioned above, $\|\boldsymbol{\zeta}\| \leq \lambda_{\min}(\mathbf{K_s})$. It follows that $\lambda_{min}(\mathbf{K_s}) - \|\boldsymbol{\zeta}\| = k_\zeta$, where $k_\zeta > 0$, and, for the minimum eigenvalue of $\mathbf{M}^{-1}$, we have $\lambda_{min}(\mathbf{M}^{-1}) = k_m$, with $k_m > 0$. Consequently, the derivative of the Lyapunov candidate in (54) is

$$\begin{aligned} \dot{V} &\leq -k_m k_\zeta \|s\| \\ &\leq -k_\alpha \|s\| \end{aligned} \tag{55}$$

with $k_m k_\zeta = k_\alpha$. Therefore, the sliding condition is fulfilled. Moreover, the derivative of the Lyapunov candidate is negative definite $\dot{V} < 0$, which means that the system is asymptotically stable, i.e., the velocity error $\tilde{\mathbf{u}} \to \mathbf{0}$ as $t \to \infty$.

Now, let us separate Equation (55) in terms of s, so that $\frac{s^T \dot{s}}{\|s\|} \leq -k_\alpha$. Then, by integrating both sides over the time interval $[0, t_a]$, we obtain

$$\int_0^{t_a} \frac{s^T \dot{s}}{\|s\|} dt \leq \int_0^{t_a} -k_\alpha dt \longrightarrow \|s(t_a)\| - \|s(0)\| \leq -k_\alpha t_a \tag{56}$$

Be reminded that $s(t_a) = 0$, i.e., with the proposed control action in t_a, the sliding surface is reached; therefore,

$$t_a \geq \frac{\|s(0)\|}{k_\alpha} \tag{57}$$

Consequently, the sliding surface s is reached in a finite time less than or equal to t_a, which depends on the initial condition of the system, perturbations, and controller gains.

4.3. Robustness Analysis

For the design of the controller, it is assumed that the estimated target velocity is equal to the actual velocity. Now, in the case where the estimated target velocity is not equal to the actual velocity, it follows that $\dot{\hat{\boldsymbol{\eta}}}_\mathbf{o} \neq \dot{\boldsymbol{\eta}}_\mathbf{o}$. Consequently, $\mathbf{J_o}\hat{\mathbf{u}}_\mathbf{o} \neq \mathbf{J_o}\mathbf{u_o}$; hence, there is an error of velocity estimation defined as $\tilde{\mathbf{u}}_\mathbf{o} = \mathbf{u_o} - \hat{\mathbf{u}}_\mathbf{o}$. By clearing the velocity estimate,

$$\hat{\mathbf{u}}_\mathbf{o}(t) = \mathbf{u_o}(t) - \tilde{\mathbf{u}}_\mathbf{o}(t) \tag{58}$$

Now, replacing the control action $\mathbf{u}(t) \approx \mathbf{u_k}(t)$ in (38) into the derivative of the Lyapunov candidate proposed previously (29),

$$\begin{aligned} \dot{V} &= s^T\Big(-\mathbf{J}\Big(\mathbf{J}^\dagger(-\dot{\hat{\boldsymbol{\eta}}}_\mathbf{o} + \boldsymbol{\lambda}\tilde{\boldsymbol{\eta}} + \mathbf{K_d} sat(s,\varepsilon)) + \Big(\mathbf{I} - \mathbf{J}^\dagger\mathbf{J}\Big)\boldsymbol{\Delta}_\mathbf{u}\Big) - \dot{\boldsymbol{\eta}}_\mathbf{o} + \boldsymbol{\lambda}\tilde{\boldsymbol{\eta}}\Big) \\ &= s^T(\dot{\hat{\boldsymbol{\eta}}}_\mathbf{o} - \boldsymbol{\lambda}\tilde{\boldsymbol{\eta}} - \mathbf{K_d} sat(s,\varepsilon) - \dot{\boldsymbol{\eta}}_\mathbf{o} + \boldsymbol{\lambda}\tilde{\boldsymbol{\eta}}) \\ &= s^T(\dot{\hat{\boldsymbol{\eta}}}_\mathbf{o} - \mathbf{K_d} sat(s,\varepsilon) - \dot{\boldsymbol{\eta}}_\mathbf{o}) \\ &= s^T(\mathbf{J_o}\hat{\mathbf{u}}_\mathbf{o} - \mathbf{K_d} sat(s,\varepsilon) - \mathbf{J_o}\mathbf{u_o}) \\ &= s^T(\mathbf{J_o}\mathbf{u_o} - \mathbf{J_o}\tilde{\mathbf{u}}_\mathbf{o} - \mathbf{K_d} sat(s,\varepsilon) - \mathbf{J_o}\mathbf{u_o}) \\ &= s^T(-\mathbf{J_o}\tilde{\mathbf{u}}_\mathbf{o} - \mathbf{K_d} sat(s,\varepsilon)) \\ &= -s^T\mathbf{J_o}\tilde{\mathbf{u}}_\mathbf{o} - s^T\mathbf{K_d} sat(s,\varepsilon) \end{aligned} \tag{59}$$

In order to obtain $\dot{V} < 0$, i.e., it is negative definite, the following sufficient condition must be fulfilled:

$$\begin{aligned} s^T\mathbf{K_d} sat(s,\varepsilon) &> s^T\mathbf{J_o}\tilde{\mathbf{u}}_\mathbf{o} \\ \left\|s^T\mathbf{K_d} sat(s,\varepsilon)\right\| &> \left\|s^T\mathbf{J_o}\tilde{\mathbf{u}}_\mathbf{o}\right\| \\ \|s\|\|\mathbf{K_d}\| &> \|s\|\|\mathbf{J_o}\tilde{\mathbf{u}}_\mathbf{o}\| \\ \lambda_{min}(\mathbf{K_d}) &> \|\mathbf{J_o}\tilde{\mathbf{u}}_\mathbf{o}\| \end{aligned} \tag{60}$$

This condition states that the gain matrix $(\mathbf{K_d})$ can be selected to be as large as desired; on the other hand, knowing the tracking velocity error boundary ($\mathbf{J_o}\tilde{\mathbf{u}}_\mathbf{o}$), the matrix $(\mathbf{K_d})$ must be larger than the boundary for the system to be asymptotically stable. In other words, if (60) is satisfied, it follows that the derivative of the Lyapunov candidate is $\dot{V} < 0$, i.e., definite negative, and the sliding condition is satisfied:

$$\begin{aligned}\dot{V} &= -\left\|s^T\mathbf{J_o}\tilde{\mathbf{u}}_\mathbf{o}\right\| - \left\|s^T\mathbf{K_d}sat(s,\varepsilon)\right\| = -\left(\left\|s^T\mathbf{J_o}\tilde{\mathbf{u}}_\mathbf{o}\right\| + \left\|s^T\mathbf{K_d}sat(s,\varepsilon)\right\|\right)\\ &\leq -(\|s\|\|\mathbf{J_o}\tilde{\mathbf{u}}_\mathbf{o}\| + \|s\|\|\mathbf{K_d}\|) = -\|s\|(\|\mathbf{J_o}\tilde{\mathbf{u}}_\mathbf{o}\| + \|\mathbf{K_d}\|)\\ &\leq -(\|\mathbf{J_o}\tilde{\mathbf{u}}_\mathbf{o}\| + \lambda_{\min}(\mathbf{K_d}))\|s\|\end{aligned} \tag{61}$$

$k_d = \lambda_{\min}(\mathbf{K_d})$ is the smallest eigenvalue of the gain matrix $\mathbf{K_d}$ and $k_{\tilde{u}} = \|\mathbf{J_o}\tilde{\mathbf{u}}_\mathbf{o}\|$ is the error norm of the target velocity estimate. It can be concluded that

$$\begin{aligned}\dot{V} &\leq -(k_{\tilde{u}} + k_d)\|s\|\\ &\leq -\alpha_k\|s\|\end{aligned} \tag{62}$$

where $\alpha_k = (k_{\tilde{u}} + k_d)$.

5. Experimental Results

In order to validate the proposed control scheme, several experiments are carried out. The first experiment consists of comparing the behaviors of the controller without dynamic compensation and with dynamic compensation under the same trajectory of the moving target, while the second experiment evaluates the behavior of the total scheme, in which the target performs a movement that generates changes in displacement and rotation. For the experimental tests, the DJI Matrice 600 Pro Drone equipped with a RealSense™ Depth Camera D455 is used. The camera is located at the bottom of the UAV, as can be seen in Figure 4. More detailed characteristics of the composition of the robot can be seen in [11]. In addition, a unicycle-type robot is used to emulate the moving target. The robot has an internal computer that acts independently of the hexarotor, for which the robot is commanded by velocities corresponding to a predefined trajectory (Experiments 1 and 2). The robot executes the velocity set in the working environment $\{\mathcal{F}\}$, where the target of interest is the main feature (red circular object) located at the top of the unicycle robot. In addition, it has two secondary features that help to determine the orientation of the robot. The orientation is obtained by the secondary characteristic points captured by the camera, by means of estimating the triangulation of the characteristics of the relative orientation of the target with respect to the UAV. All this processing is included in the UAV's on-board computer. Finally, a Kalman filter is used to estimate the target velocity, as detailed in [30].

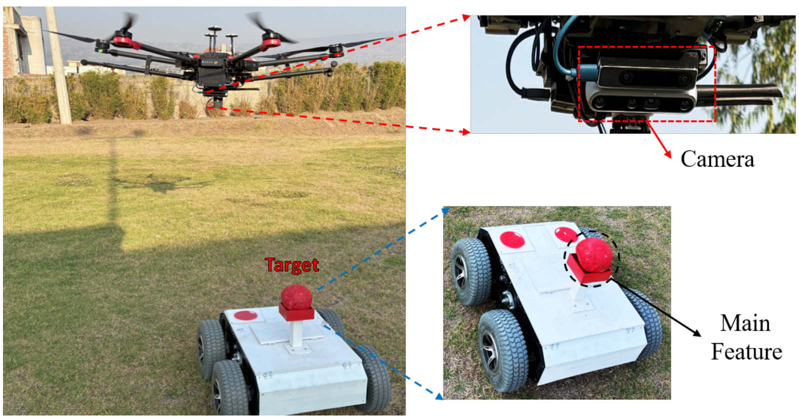


Figure 4. Robots for experiments. At the bottom is the target located on the unicycle robot with two marks to obtain the orientation of the target, while the hexarotor with the built-in vision system can be seen at the top.

5.1. Dynamic Model Validation

In Figure 5, the validation of the dynamic model is presented, where the robot was excited with a reference signal u_{ref} and the real velocity u_l is obtained. In addition, depending on the velocity u_l, the velocity is calculated by the simplified dynamic model. It is observed that the signal of the real robot is similar to that of the model.

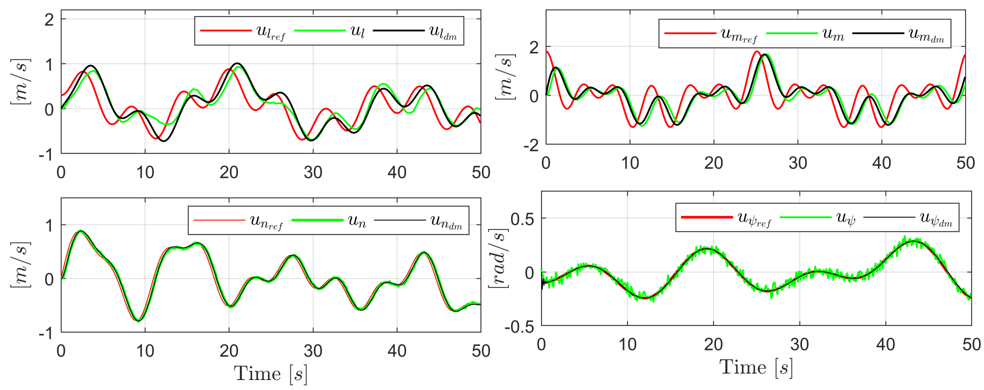


Figure 5. Dynamic model validation of the Matrice 600PRO UAV. The red line represents the reference signal, the green line represents the actual velocity obtained from the UAV, and the black line represents the signal obtained from the dynamic model.

Now, the parameters considered in the control scheme for experiments are detailed in Table 1.

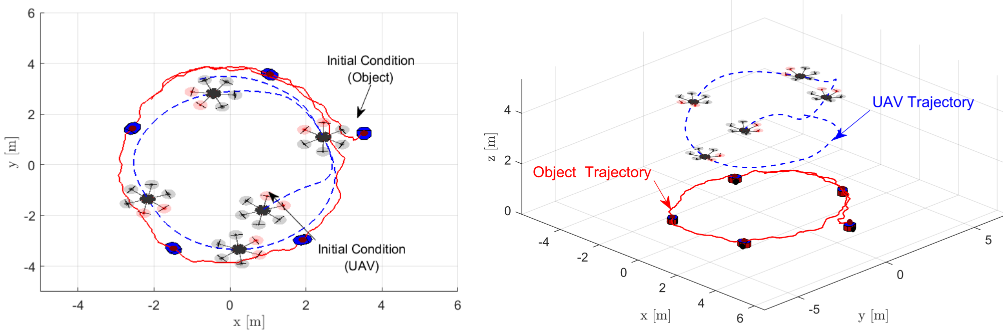
Table 1. Controller parameters.

Parameter	Parameter
$\mathbf{K_n} = diag([0.5 \quad 0.75])$	$\mathbf{K_u} = diag([0.1 \quad 0.1 \quad 0.05 \quad 0.1])$
$\mathbf{K_d} = diag([7.5 \quad 10])$	$\mathbf{K_s} = diag([0.2 \quad 0.2 \quad 0.2 \quad 0.2])$
$\epsilon = 0.01 \quad \sigma = 0.75 \quad k_\psi = 1.2$	$\boldsymbol{\eta}_\mathbf{d} = [0, 0]^T$[pixel]

5.2. Experiment 1

In this experiment, the target follows a circular trajectory. Two tests are performed: in the first one, the tracking of the target is performed without dynamic compensation; the second one is with a dynamic compensator. The initial position of the UAV is $\boldsymbol{\xi}_\mathbf{p}(t_0) = [1 \quad -2 \quad 4]^T$ [m]; the initial orientation is $\xi_\psi(t_0) = \frac{\pi}{4}$ [rad]; and the safe following distance is $\rho_d = 2$ [m]. The experiment is carried out until $t_f = 60$ [s]. Figure 6 shows the movement of the UAV and the target with the dynamic compensation controller. It can be seen that the UAV follows the target at a safe distance.

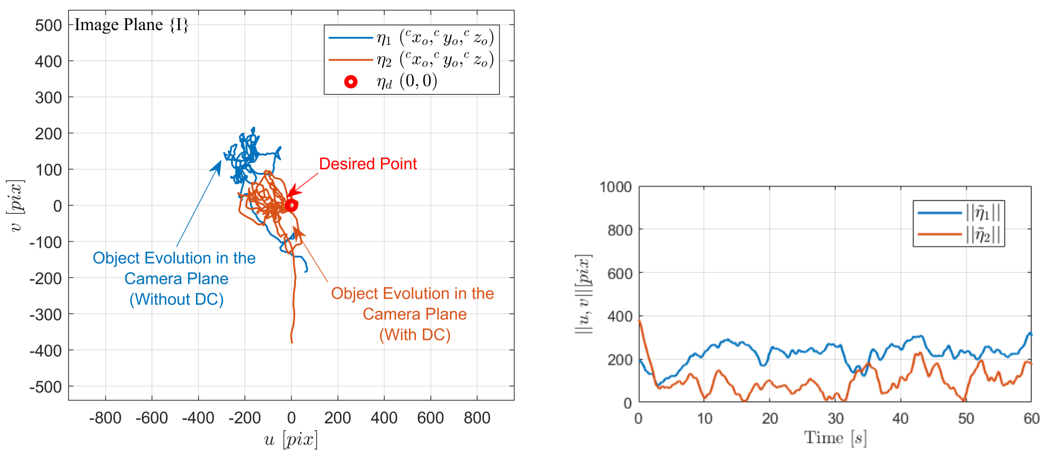
In Figure 7a, the behavior of the target in the image plane is presented. The blue line represents the evolution of the target without dynamic compensation, while the orange line represents the evolution of the target with the full control scheme (VSC-SM + CD-SMC). Figure 7b presents the Euclidean norm of the servo visual control error in the servo-visual control $\|\boldsymbol{\eta}\|(t)$. It is evident that the controller with dynamic compensation performs better than the controller without dynamic compensation, approximating the system states to the desired conditions. However, the performance of the servo visual controller based on sliding mode (VSC-SM) tries to keep the target in the desired position on the image plane while preventing the target leaving the robot's field of view (FOV).



(**a**) XZ plane view. (**b**) View in XYZ space.

Figure 6. Stroboscopic movement of the UAV during Experiment 1 in $\{\mathcal{F}\}$: (**a**) represents the movement in the XY plane, and (**b**) represents the stroboscopic movement of the robot in the XYZ space.

Finally, Figure 8 shows the evolution of the secondary task. It can be seen that the error of the UAV distance ($\tilde{\rho}(t)$) with respect to the target with (VSC-SM + DC-SMC) compared to the control (VSC-SM) is lower. In the same way, it can be seen that the orientation error behaves with both controllers. Note that, even though they are secondary tasks, it is observed that the error approaches or stays close to zero. This ensures that the UAV follows at a safe distance and in the same direction of movement as the target.



(**a**) Target behavior in the image plane $\{\mathcal{I}\}$. (**b**) VSC-SM error module.

Figure 7. Target behavior in the image plane of the controller without dynamic compensation (blue line) and the controller with dynamic compensation (orange line).

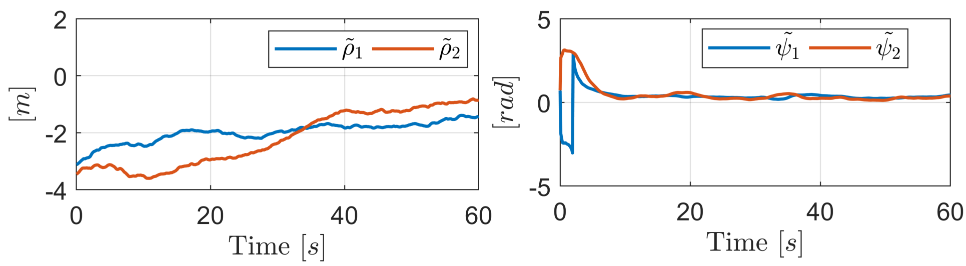
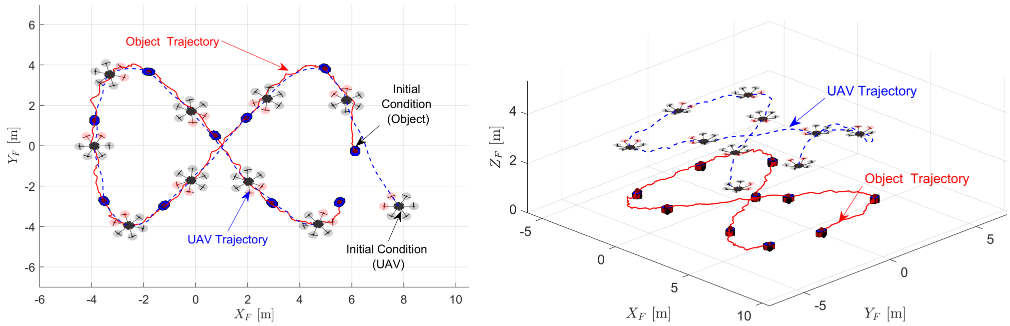


Figure 8. Secondary objective error.

5.3. Experiment 2

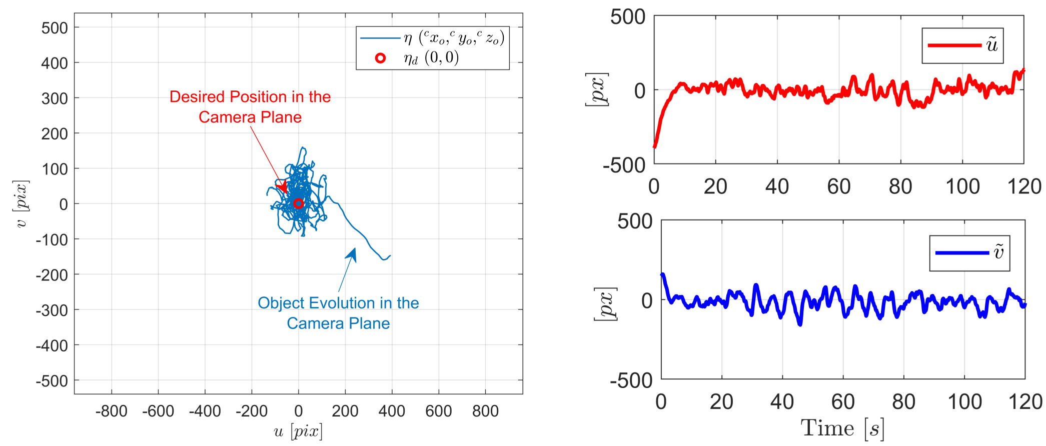
For this experiment, the mobile robot executes more pronounced movements on the floor. The initial position of the UAV is $\boldsymbol{\xi}_\mathbf{p}(t_0) = \begin{bmatrix} 8 & -3 & 4 \end{bmatrix}^T$ [m]; the initial orientation is $\zeta_\psi(t_0) = \frac{5}{4}\pi$ [rad]; and the safe following distance is $\rho_d = 2.5$ [m]. The experiment is carried out until $t_f = 120$ [s]. Figure 9 presents the movement of the robot performed in the workspace $\{\mathcal{F}\}$; it can be seen that the evolution of the UAV trajectory follows the trajectory of the moving target, keeping the desired distance and maintaining the rotation of the target.



(**a**) XY-plane view. (**b**) XYZ-space view.

Figure 9. Stroboscopic movement executed by the UAV during the experiment.

Figure 10a shows the evolution of the target in the image plane $\{\mathcal{I}\}$. It can be seen that it keeps the target in the center of the image plane, i.e., in the desired position. In Figure 10b, we can see the control errors ($\tilde{\eta}(t)$) produced in the experiment. It is clear that the errors are close to zero, validating the performance of the controller (VSC-SM). This verifies that the main task, which is the tracking of the moving target, is fulfilled without any problem.



(**a**) Target evolution in $\{\mathcal{I}\}$. (**b**) Control error $\tilde{\boldsymbol{\eta}}$.

Figure 10. Evolution of control errors of principal tracking task and target behavior in the image plane during Experiment 2.

Now, in Figure 11, the error of the UAV's distance to the target during the experimental test is shown. And the figure shows the error of the UAV orientation with respect to the target. It is clear that these control objectives are achieved in the best way, due to the redundancy of the system (Equation (21)).

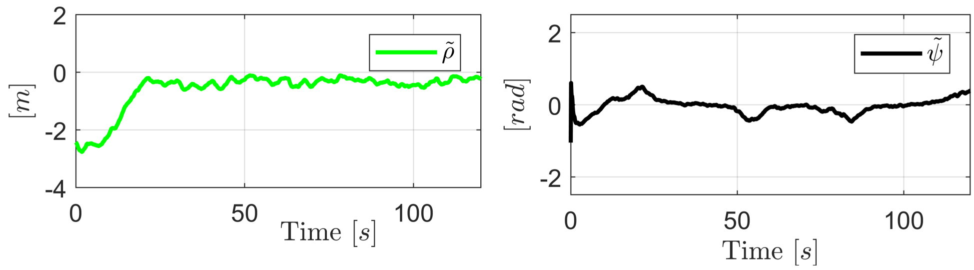


Figure 11. Secondary task error evolution.

The target's estimated velocity is presented in Figure 12. It can be seen that the speed of the target estimated by measuring the position of the target with respect to the camera is similar to the real speed of the target, which allows the UAV to not lose sight of the target and to follow it. With this, it is possible to verify the robustness of the controller, which, although it has an error in the velocity estimation, the UAV does not lose sight of the target and follows it during the whole experiment.

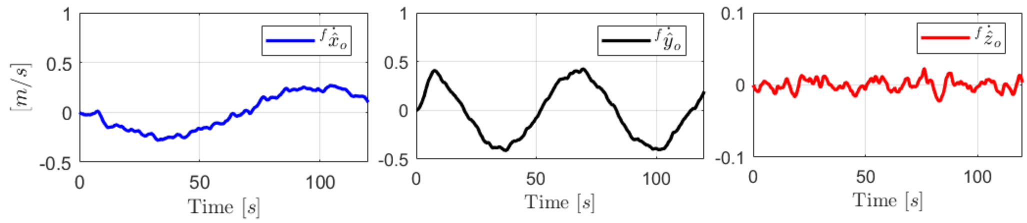


Figure 12. Target velocity estimated by [30].

Figure 13 shows the maneuverability velocities generated by the dynamic compensator based on sliding mode. It can be seen that these velocities contain small oscillations induced by the discontinuous part of the controller.

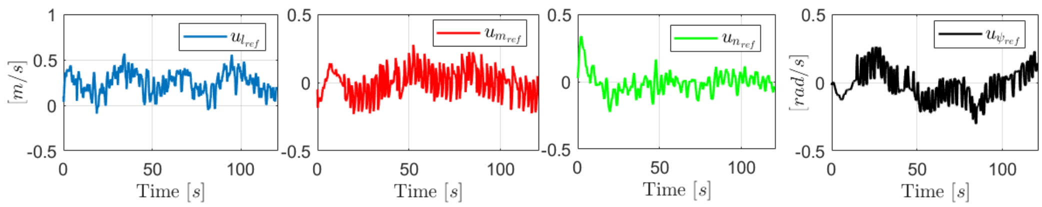


Figure 13. Reference velocities generated by DC-SMC in Experiment 2.

Finally, the maneuverability velocity errors are presented in Figure 14. The errors are very close to zero, due to the dynamic compensation control (Equation (52)).

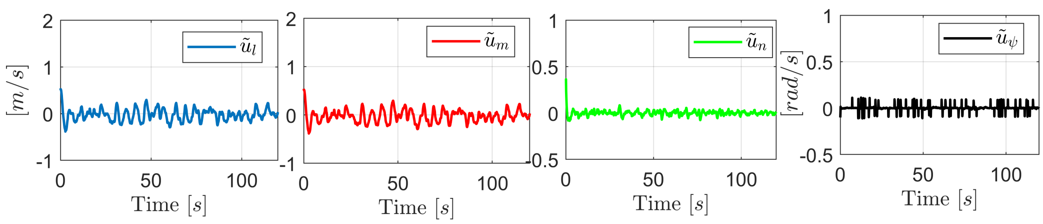


Figure 14. UAV maneuverability velocity errors.

6. Conclusions

A visual servo control scheme for the tracking of moving targets based on the sliding-mode technique is presented as the primary task for a UAV and, by exploiting the redundancy of the system, a secondary task is defined to track the target at a safe distance and oriented in the direction of the moving target. Experiment 1 validates the operation of the VSC-SM only, keeping the target of interest within the field of view (FOV) of the UAV. Moreover, since the experimental tests are performed in an uncontrolled environment, the disturbances induced by the wind and by errors induced by the measurement of the target position, i.e., these disturbances affect the controller and the UAV does not lose track of the target of interest. All this shows that the non-linear controller is stably deduced in the stability analysis and by means of the robustness analysis under external disturbances.

On the other hand, the full scheme with the dynamic compensation block based on the sliding-mode technique (DC-SMC) is also successfully used to reduce the velocity errors. The results of Experiment 1 and Experiment 2 show the behavior of the controller. The tracking performance of the target is better and it is able to reduce the control errors. Similarly, as the experiment is a non-controlling environment (non-smooth floor), there are several disturbances, and, in addition to being an approximate dynamic model that contemplates model errors, the controller keeps the velocity errors very close to zero, thus experimentally validating the DC-SMC.

The proposed control strategy proves to be robust for the tracking of moving targets and, being a redundant system, it is possible to accomplish secondary tasks without affecting the main task. For future work, we can consider adding a third task, which is obstacle avoidance during target tracking. Finally, the algorithm for velocity estimation [30] shows good results in real experiments.

For future work, it is planned to implement the proposed controller based on sliding mode for object grasping using a robot consisting of a drone and a robotic arm on the bottom of the drone.

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Appendix A

Table A1. Nomenclature used in the article.

Symbol	Description	Symbol	Description
$\{\mathcal{F}\}$	Fixed reference system	$\{\mathcal{A}\}$	UAV mobile reference system

Table A1. *Cont.*

Symbol	Description	Symbol	Description
$\{\mathcal{C}\}$	Camera reference system in the UAV	$\{\mathcal{I}\}$	Image plane reference system
$\boldsymbol{\xi}_{\mathbf{p}}$	UAV centre of mass position vector	ξ_ψ	UAV operation point orientation
$\boldsymbol{\xi}$	UAV operating point position and orientation vector	$\dot{\boldsymbol{\xi}}_{\mathbf{p}}$	UAV centre of mass velocity vector
$\dot{\xi}_\psi$	UAV rotational velocity	$\dot{\boldsymbol{\xi}}$	UAV operating point velocity vector
$\boldsymbol{\xi}^*$	Differential kinematics of the proposed secondary objectives	$\mathbf{R}(x,\phi)$	Rotation matrix around the X-axis at an angle ϕ
${}^b\mathbf{R}_a$	Rotation matrix of the reference frame a with respect to b	$\mathbf{I}_{m\times n}$	Matrix identity of dimension $m \times n$
$\mathbf{0}_{m\times n}$	Null matrix of dimension $m \times n$	$\mathbf{u}$	Current UAV velocity vector
u_l	Frontal linear velocity of UAV	u_m	Lateral linear velocity of UAV
u_n	UAV linear rate of elevation	u_ψ	UAV rotational velocity
$\mathbf{J_A}(\psi)$	UAV Jacobian matrix	$\mathbf{M}, \mathbf{C}$	Dynamic UAV model matrices
$\mathbf{u_{ref}}$	UAV reference velocity vector from the dynamic model	$u_{l_{ref}}$	Front velocity reference
$u_{m_{ref}}$	Lateral velocity reference	$u_{n_{ref}}$	Reference of elevation velocity
$u_{\psi_{ref}}$	Angular velocity reference	$\dot{\mathbf{u}}$	Vector of UAV acceleration
$\dot{u}_l$	Forward acceleration	$\dot{u}_m$	Lateral acceleration
$\dot{u}_n$	Acceleration of elevation	$\dot{u}_\psi$	Angular acceleration
δ	Parameter vector of the UAV dynamic model	${}^{\mathbf{b}}\mathbf{p_a}$	Position of a vector in the system a with respect to the system b
${}^{\mathbf{b}}\dot{\mathbf{p}}_{\mathbf{a}}$	Velocity of a vector in the system a with respect to the system b	$\boldsymbol{\eta}$	Vector of target position in the image plane
u	Position of the object in the image plane on the horizontal axis	v	Position of the object in the image planeoin the vertical axis
f	Focal length of the camera	$\dot{\boldsymbol{\eta}}$	Target velocity vector in the image plane
${}^{\mathbf{f}}\mathbf{v_c}$	Linear velocity of the camera relative to the fixed reference frame	${}^{\mathbf{f}}\boldsymbol{\omega}_{\mathbf{c}}$	Angular velocity of the camera relative to the fixed reference frame
$\mathbf{T_A}$	Transformation matrix	$\mathbf{J_o}$	Jacobian target matrix
$\mathbf{J}$	Jacobian matrix of target interaction	$\mathbf{J_n}$	Jacobian matrix of the secondary objectives
$\dot{\boldsymbol{\eta}}_{\mathbf{o}}$	Target velocity in the fixed system	$\tilde{\boldsymbol{\eta}}$	Tracking error of the target
$\dot{\tilde{\boldsymbol{\eta}}}$	Tracking error velocity	s	Sliding surface
$\dot{s}$	First derivative of sliding surface	$\mathbf{K_n}$	Weight matrix of continuous controller action for tracking
$\mathbf{K_d}$	Weight matrix of discontinuous controller action for tracking	$\mathbf{u_k}$	Vector kinematic controller for target tracking
$\mathbf{u_c}$	Continuous control action of the target tracking controller	$\mathbf{u_d}$	Discontinuous control action of the target tracking controller
$\mathbf{u_n}$	Control action of the secondary objectives	$\lambda_{min}(X)$	Minimum eigenvalue of a matrix X
$\lVert s \rVert$	Norm of vector s	$\tanh(\bullet)$	Hyperbolic tangent
$\boldsymbol{\xi}^*_d$	Secondary task error vector	$\hat{\psi}_o$	Estimated velocity of the target
$\tilde{\rho}$	Safe distance error	ρ_d	Safe distance desired
$\tilde{\xi}_\psi$	Orientation error	$\mathbf{K_o}$	Matrix weighting secondary task errors
$\mathbf{K_u}$	Matrix of weights of the continuous part of the dynamic compensation	$\mathbf{K_s}$	Matrix of weights of the discontinuous part of the dynamic compensation
$\tilde{\mathbf{u}}$	Velocity error vector	$\dot{\tilde{\mathbf{u}}}$	Derivative of the velocity error vector
ζ	Vector of disturbances and/or errors in dynamic modeling	$\mathbf{v_{ref}}$	Control action of dynamic compensation

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